

BASIC MACHINE LEARNING

Session I

Foundations • Evaluation Metrics • Linear & Logistic Regression

A Beginner-Friendly Introduction to Machine Learning



What We Will Cover Today

1

Introduction to Machine Learning

3

Supervised Learning & Model Evaluation Metrics

5

Logistic Regression

2

Data Preprocessing & Train-Test Split

4

Linear Regression

6

Hands-on Practicals After Every Concept

By The End Of This Session, You Will Be Able To

- Explain what Machine Learning is and its core types — supervised, unsupervised, and reinforcement learning
- Recall data preprocessing essentials and perform a train-test split
- Interpret classification metrics (accuracy, precision, recall, F1, confusion matrix) and regression metrics (RMSE, R^2)
- Build and interpret a Linear Regression model
- Build and interpret a Logistic Regression model
- Classify data using K-Nearest Neighbors (KNN) and choose a sensible value of K
- Understand SVM and when to prefer it over other classifiers

WHY IT MATTERS

These foundations power prediction, classification, and decisioning inside modern AI and Agentic AI systems.

PART 1 · FOUNDATIONS

Introduction to Machine Learning



What is Machine Learning?

Teaching computers to learn from examples, instead of hard-coded rules

Definition

A system learns patterns from data, and uses them to make predictions or decisions — without being explicitly programmed for every scenario.

Everyday Analogy

A child learns to recognize a dog *not* from a rulebook of features, but by seeing many labeled examples — then recognizing a dog they've never seen before.

Real-world example: Spam filter

Trained on emails labeled spam / not-spam, it learns word and sender patterns, then classifies brand-new emails on its own.

Traditional Programming vs. Machine Learning

Traditional Programming	Machine Learning
You write the rules	The computer learns the rules
Input + Rules → Output	Input + Output → Rules (model)
Good for fixed logic (e.g., a calculator)	Good for complex patterns (e.g., spam detection)

Takeaway: Learning patterns from data to predict — instead of following hardcoded instructions.



SUMMARY

Traditional Programming:
DATA + RULES → OUTPUT

Machine Learning:
DATA + OUTPUT (examples) → RULES (MODEL)

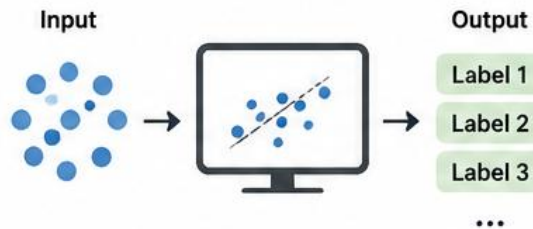
TYPES OF MACHINE LEARNING



SUPERVISED LEARNING

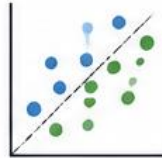
Learns from labeled data

How it works



Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines (SVM)
- Random Forest



Examples



House Price Prediction



Email Spam Detection



Disease Prediction



UNSUPERVISED LEARNING

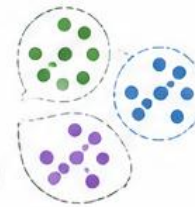
Finds patterns in unlabeled data

How it works



Common Algorithms

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Principal Component Analysis (PCA)
- Association Rule Learning



Examples



Customer Segmentation



Market Basket Analysis



Anomaly Detection



REINFORCEMENT LEARNING

Learns by interacting with an environment

How it works



Common Algorithms

- Q-Learning
- Deep Q-Networks (DQN)
- Policy Gradient
- Actor-Critic Methods
- Proximal Policy Optimization (PPO)



Examples



Game Playing (Atari, Chess)



Robotics Control

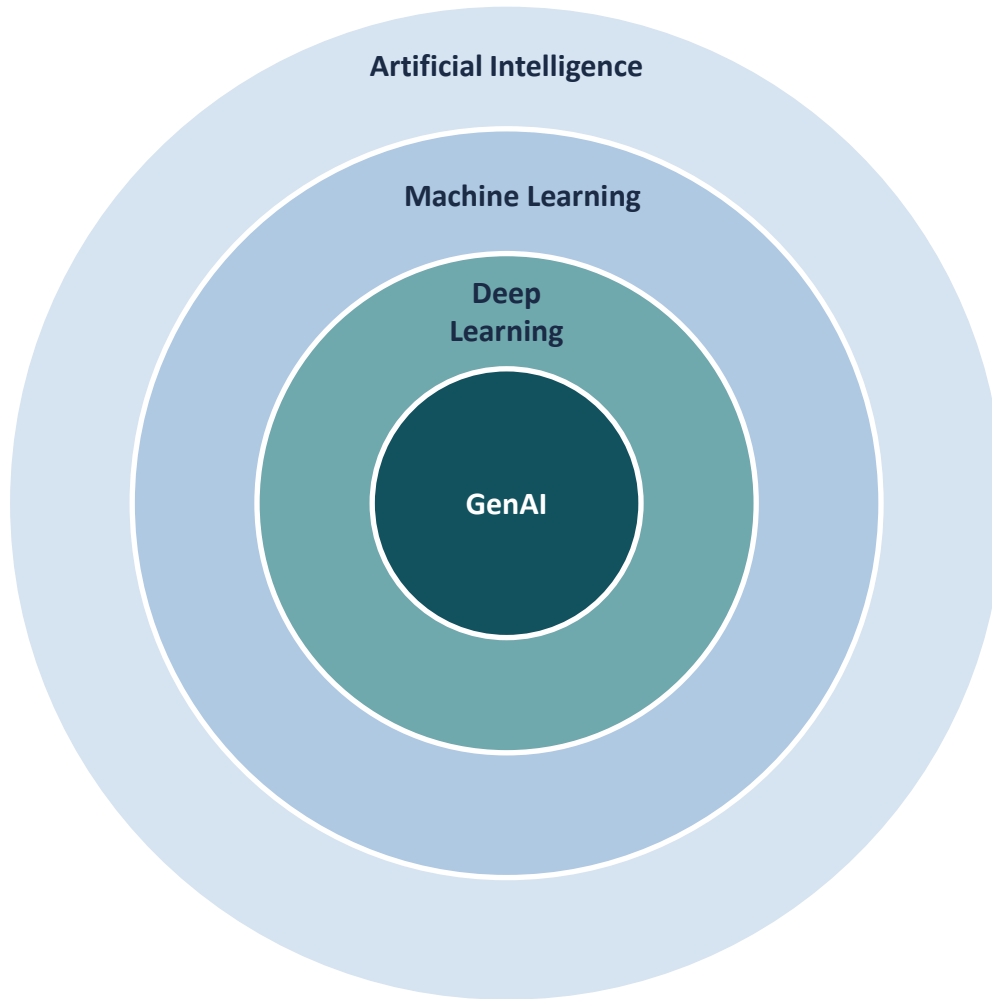


Autonomous Driving



Summary: **Supervised Learning** learns from labeled data, **Unsupervised Learning** finds hidden patterns in data, **Reinforcement Learning** learns by trial and error to maximize rewards.

Where Machine Learning Fits



- AI is the broad goal of building intelligent systems.
- ML is a subset of AI that learns patterns from data.
- Deep Learning uses neural networks with many layers.
- GenAI (built on Deep Learning) generates new content — text, images, and more.

Supervised vs. unsupervised learning methods

How labeled data separates the two learning paradigms

		Label present?		Type of label	
Supervised learning method	Regression	-----	✓	Score of a particular student in predictive analytics course	Continuous
	Classification	-----	✓	Spam or ham for past email data	Categorical
Unsupervised learning method	Clustering	-----	✗	No labeled outcome — the model finds structure on its own	

Supervised Learning: Learning From Labeled Examples

In supervised learning, every training example comes with the correct answer. The model's job is to learn the mapping from input to output, then apply it to new, unseen inputs.

Regression

Predicts a continuous number.

Example: predicting a house's sale price, or tomorrow's temperature.

Classification

Predicts a category (a class label).

Example: predicting whether a tumor is malignant or benign, or whether an email is spam.

Today's session covers one regression algorithm (Linear Regression) and three classification algorithms (Logistic Regression, KNN, and SVM).

PART 1 · FOUNDATIONS

Data Preparation Recap

A quick refresher before we build models



Preparing Data Before Modeling

1

Handle Missing Values

Fill or remove blanks so the model does not break on incomplete rows.

2

Encode Categorical Data

Convert text categories (e.g. city, gender) into numbers the model can use.

3

Scale Numeric Features

Bring features onto a similar scale so no single feature dominates the model.

Note: A deep-dive into data cleaning and pipelines is covered in the Data Engineering for AI Applications session. Here we recap only what is needed to move forward.

DATA PREPROCESSING IN ML



Data preprocessing is the process of transforming raw data into a **clean, consistent and structured** format that is suitable for machine learning models.

WHY IS IT IMPORTANT?

- ✓ Improves data quality
- ✓ Helps models learn better patterns
- ✓ Reduces training time
- ✓ Increases model accuracy and reliability
- ✓ Prevents misleading results

KEY PREPROCESSING STEPS



1. DATA CLEANING

- Handle missing values (mean/median/mode, forward fill)
- Remove duplicates
- Fix inconsistent or incorrect data



2. DATA TRANSFORMATION

- Normalize (Min-Max Scaling)
- Standardize (Z-Score Scaling)
- Log/Box-Cox transformation



3. ENCODING CATEGORICAL DATA

- Label Encoding (ordinal data)
- One-Hot Encoding (nominal data)
- Target / Frequency Encoding



4. FEATURE ENGINEERING

- Create new features
- Combine existing features
- Extract meaningful information



5. OUTLIER HANDLING

- Detect outliers (IQR, Z-Score)
- Remove or cap outliers

RAW DATA (EXAMPLE)

Age	Salary	City	Purch.
25	50000	Delhi	No
?	60000	Mumbai	Yes
30		Delhi	No
25	50000	Delhi	No
35	120000	?	Yes

AFTER PREPROCESSING

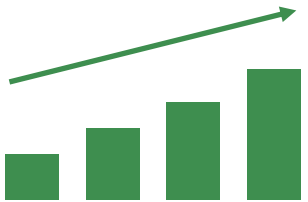
Age	Salary	City	Purch.
25	50000	Delhi	0
28	60000	Mumbai	1
30	85000	Delhi	0
25	50000	Delhi	0
35	120000	Bangalore	1

MODEL READY DATA

-1.2	0.3	1	0
-0.1	0.8	2	1
0.5	0.0	1	0
-1.2	-0.5	1	0
1.1	1.7	3	1

(Numerical & encoded format, ready for ML models)

BETTER MODEL PERFORMANCE



Higher accuracy, better generalization



Remember: Good data in → Good model out.



GOAL: Clean, consistent and meaningful data leads to accurate insights and reliable ML models.

Train-Test Split

Before training, we divide our dataset into two parts — so we can check the model on data it has never seen.

Training Set — 80%

**Test
20%**

Learning Part

The model studies this data to learn patterns and relationships.

Checking Part

Held back and used only to evaluate how well the model generalizes.

HANDS-ON · PRACTICAL 1

Preprocessing, Split & Baseline Metric

Open the notebook: recap preprocessing, split the data, and compute a baseline evaluation metric.

HANDS-ON PRACTICAL



PART 1 · FOUNDATIONS

Model Evaluation Metrics

How do we know if a model is any good?



The Confusion Matrix

A confusion matrix compares what the model predicted against what actually happened.



Classification Metrics I: Accuracy & Precision

Accuracy

FORMULA

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

RANGE OF VALUES

❖ 0 to 1 (often shown as 0% to 100%). Higher is better.

PROS

- Simple and easy to explain to anyone
- Works well when classes are roughly balanced

CONS

- Misleading on imbalanced data
- Does not show which class is being missed

Precision

FORMULA

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

RANGE OF VALUES

❖ 0 to 1 Higher means fewer false alarms.

PROS

- Useful when a false positive is costly
- Shows the quality of positive predictions

CONS

- Ignores false negatives entirely
- Can look high even if many positives are missed

Classification Metrics II: Recall & F1 Score

Recall (Sensitivity)

FORMULA

$$Recall = \frac{TP}{(TP + FN)}$$

RANGE OF VALUES

❖ 0 to 1. Higher means fewer missed cases.

PROS

- Useful when missing a positive case is costly
- Shows how completely actual positives are caught

CONS

- Ignores false positives entirely
- Can look high even with many false alarms

F1 Score

FORMULA

$$F1 = 2 \times \frac{Precision \times Recall}{(Precision + Recall)}$$

RANGE OF VALUES

❖ 0 to 1. Higher balances precision and recall.

PROS

- Balances precision and recall in one score
- Especially useful for imbalanced classes

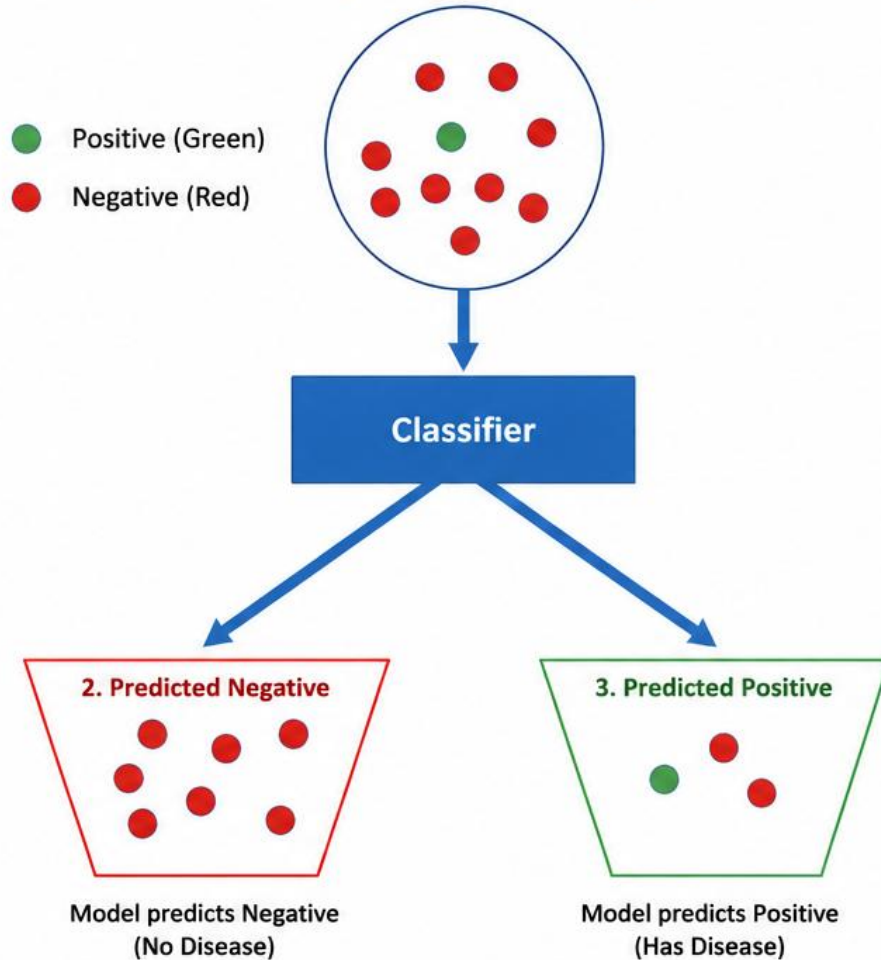
CONS

- Less intuitive to explain directly
- Treats precision and recall as equally important

Classification Metrics – Easy Example

Let's understand Accuracy, Precision, Recall and F1 Score using a simple example.

1. Population (Real World)



4. Confusion Matrix (Results)

		Real (Actual)	
		Positive (Has Disease)	Negative (No Disease)
Predicted	Positive (Has Disease)	1 TP (True Positive)	2 FP (False Positive)
	Negative (No Disease)	0 FN (False Negative)	7 TN (True Negative)

5. Metrics Calculation

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{1 + 7}{1 + 7 + 2 + 0} = \frac{8}{10} = 80\%$$

Out of 10 cases, the model predicted 8 correctly.

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{1}{1 + 2} = \frac{1}{3} = 33.3\%$$

Of the 3 predicted positives, only 1 is actually positive.

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{1}{1 + 0} = \frac{1}{1} = 100\%$$

The model found all actual positives (no positives missed).

$$\begin{aligned} \text{F1 Score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= 2 \times \frac{0.333 \times 1.000}{0.333 + 1.000} = 2 \times \frac{0.333}{1.333} = 0.50 = 50\% \end{aligned}$$

Balances Precision and Recall in a single score.



Key Takeaway

This model is good at finding all positives (Recall = 100%), but makes some wrong positive predictions (Low Precision = 33.3%), which lowers the F1 Score.

Regression Metrics

RMSE (Root Mean Squared Error)

FORMULA

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RANGE OF VALUES

❖ 0 to ∞ , in the same units as the target. Lower is better; 0 means perfect predictions.

PROS

- Same units as the target, easy to interpret
- Penalizes large errors more heavily than small ones

CONS

- Sensitive to outliers
- Cannot be compared across different targets or scales

R² (R-Squared)

FORMULA

$$R^2 = 1 - \left(\frac{\text{Sum of Squared Errors}}{\text{Total Variance}} \right) = 1 - \frac{RSS}{TSS}$$

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

RANGE OF VALUES

❖ Typically 0 to 1 (can go negative for a very poor model). 1 means a perfect fit.

PROS

- Easy to read as “percent of variance explained”
- Scale-independent, good for comparing models

CONS

- Can increase just by adding irrelevant features
- Does not show if predictions are biased

REGRESSION METRICS: RMSE & R²

1. RMSE (Root Mean Squared Error)

Measures the average prediction error.
Lower value is better (ideal = 0).

FORMULA

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

y_i = Actual value
 $\hat{y}_i$ = Predicted value
 n = Number of observations

EXAMPLE

i	Actual (y_i)	Predicted ($\hat{y}_i$)	Error ($y_i - \hat{y}_i$)	Squared Error ($(y_i - \hat{y}_i)^2$)
1	10	9	1	1
2	15	16	-1	1
3	20	18	2	4

$$\text{MSE} = \frac{1 + 1 + 4}{3} = 2$$

$$\text{RMSE} = \sqrt{2} = 1.414$$

INTERPRETATION



On average, the predictions are about **1.414** units away from the actual values.

2. R² (Coefficient of Determination)

Measures how well the model explains the variation in the target variable. Higher value is better (ideal = 1).

FORMULA

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

RSS = Residual Sum of Squares
TSS = Total Sum of Squares
 $\bar{y}$ = Mean of actual values

EXAMPLE (Using same data)

Actual values (y): 10, 15, 20
Predicted values ($\hat{y}$): 9, 16, 18

Mean of actual values ($\bar{y}$):
(10 + 15 + 20) / 3 = 15

RSS (Residual Sum of Squares):
(10-9)² + (15-16)² + (20-18)²
= 1 + 1 + 4 = 6

TSS (Total Sum of Squares):
(10-15)² + (15-15)² + (20-15)²
= 25 + 0 + 25 = 50

CALCULATION

$$\begin{aligned} R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} \\ &= 1 - \frac{6}{50} \\ &= 1 - 0.12 \end{aligned}$$

$$R^2 = 0.88$$

INTERPRETATION



The model explains **88%** of the variation in the actual values. The remaining **12%** is not explained by the model.

QUICK SUMMARY

RMSE

- Tells us the average size of the prediction error.
- Lower is better (0 is perfect).

R²

- Tells us how well the model explains the data.
- Higher is better (1 is perfect).



PART 2 · REGRESSION

Linear Regression



Equation & Trade-offs

PREDICTION EQUATION

$$\hat{y} = \beta_0 + \beta_1 x$$

Predicted Output = Intercept + (Slope × Input)

COST FUNCTION (MSE)

$$J = \text{Average of (Actual - Predicted)}^2$$

Range of predictions: $-\infty$ to ∞ (continuous, unbounded).

PROS

- Simple, fast to train, and easy to interpret
- Works well when the relationship is roughly linear
- Coefficients directly show each feature's effect

CONS

- Assumes a linear relationship between input and output
- Sensitive to outliers
- Struggles with complex, non-linear patterns



LINEAR REGRESSION

A fundamental **Supervised Learning** algorithm used to model the relationship between a dependent variable (target) and one or more independent variables (predictors) by fitting a **linear equation** to the observed data.

1 What is Linear Regression?

Linear Regression models the relationship between a **dependent (target)** variable and one or more **independent (predictor)** variables using a linear equation.

2 Model / Equation

Simple Linear Regression

$\hat{y}$: Predicted value (dependent variable)

y : Actual value

x : Independent variable (feature)

β_1 : **Slope** (coefficient)

β_0 : **Intercept**

$$\hat{y} = \beta_0 + \beta_1 x$$

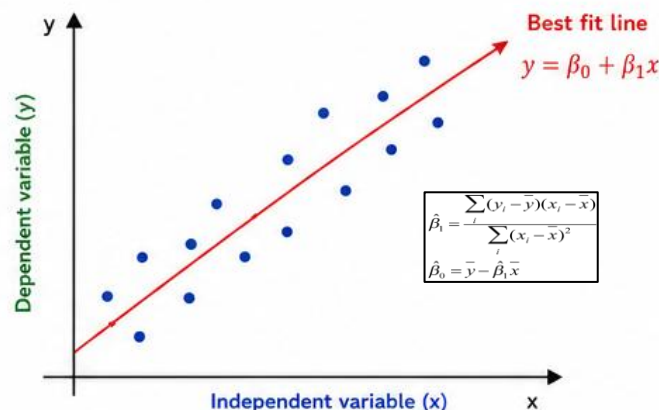
Multiple Linear Regression (General Form)

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

β_0 : Intercept $\beta_1, \beta_2, \dots, \beta_n$: Coefficients

$x_1, x_2, \dots, x_n$: Independent variables

3 Visualization (Best Fit Line)



The red line is the best fit line that minimizes the error between actual values and predicted values.

4 Cost Function (MSE)

$$J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

y_i : Actual value

$\hat{y}_i$: Predicted value

n : Number of observations

Goal : Find β_0 and β_1 that minimize $J(\beta_0, \beta_1)$

6 Evaluation Metrics

- MSE (Mean Squared Error) = $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- MAE (Mean Absolute Error) = $\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
- R^2 Score (Coefficient of Determination)

$$= 1 - \frac{SS_{res}}{SS_{tot}}$$

where,

$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{Residual Sum of Squares})$$

$$SS_{tot} = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (\text{Total Sum of Squares})$$

5 Assumptions of Linear Regression



Linearity: Relationship between X and Y is linear.



Independence of errors: Observations are independent.



Homoscedasticity: Constant variance of errors.



Normality: Errors are normally distributed.



No multicollinearity: Independent variables are not highly correlated.

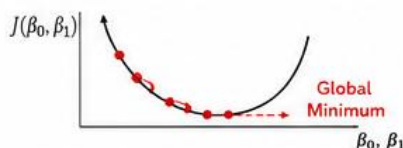
8 How Linear Regression Learns? (Gradient Descent)

We use Gradient Descent to minimize the cost function $J(\beta_0, \beta_1)$ by updating β_0 and β_1 .

$$\beta_1 := \beta_1 - \alpha \frac{\partial J}{\partial \beta_1}$$

$$\beta_0 := \beta_0 - \alpha \frac{\partial J}{\partial \beta_0}$$

α : Learning rate (controls step size)
Repeat until cost $J(\beta_0, \beta_1)$ is minimized.



7 Example

Predict house price (y) based on area (x in sq.ft).
Larger area $\rightarrow$ Higher price (linear relationship)

Area (x)	Price (y)
500	1.8
750	2.6
1000	3.4
1250	4.2
1500	5.1
...	...

We fit a line that best represents this trend.

9 Pros and Cons



- ✓ Simple and easy to understand
- ✓ Works well with linear relationships
- ✓ Fast to train and interpret
- ✓ Requires less data



- ✗ Assumes linear relationship
- ✗ Sensitive to outliers
- ✗ May underfit complex data
- ✗ Does not capture non-linear patterns

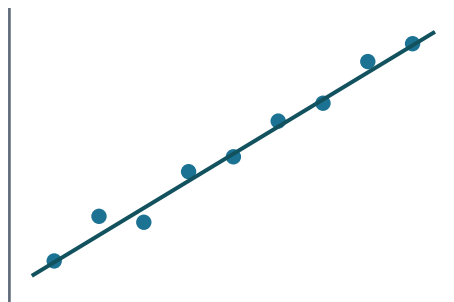


Note: Linear Regression works best when the relationship between variables is linear.
For non-linear data, use **Polynomial Regression** or other advanced models.

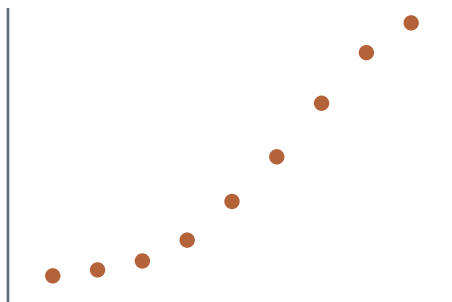
Assumptions I: Linearity & Normality

LINEARITY

X and Y should show a linear relationship — otherwise, there is no use fitting a straight line.



Linear — good fit

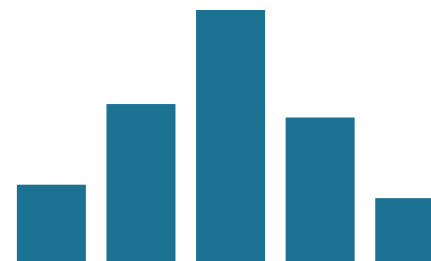


Non-linear — poor fit

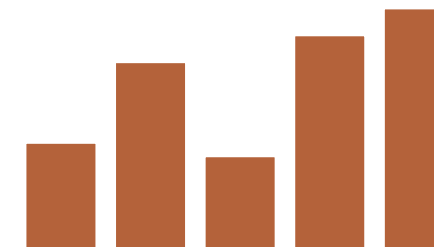
If the scatter clearly curves, a straight line will systematically under- or over-predict — consider a polynomial or non-linear model instead.

NORMALITY

Error terms (not X or Y) should be normally distributed with mean zero. This matters for trusting p-values, not for fitting the line itself.



Normally distributed



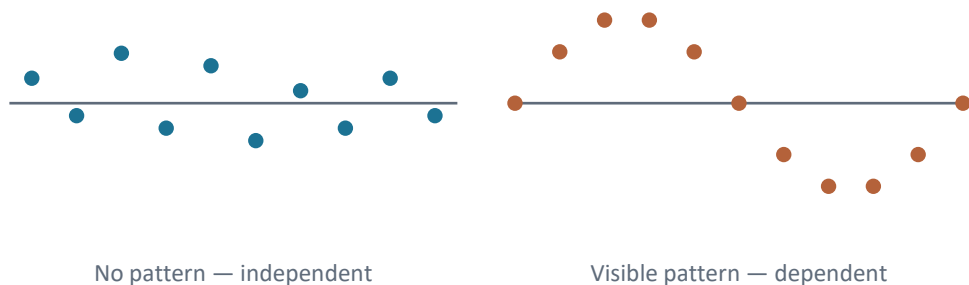
Not normally distributed

Non-normal errors do not stop you from fitting a line, but they make the p-values used to test significance unreliable.

Assumptions II: Independence & Homoscedasticity

INDEPENDENCE

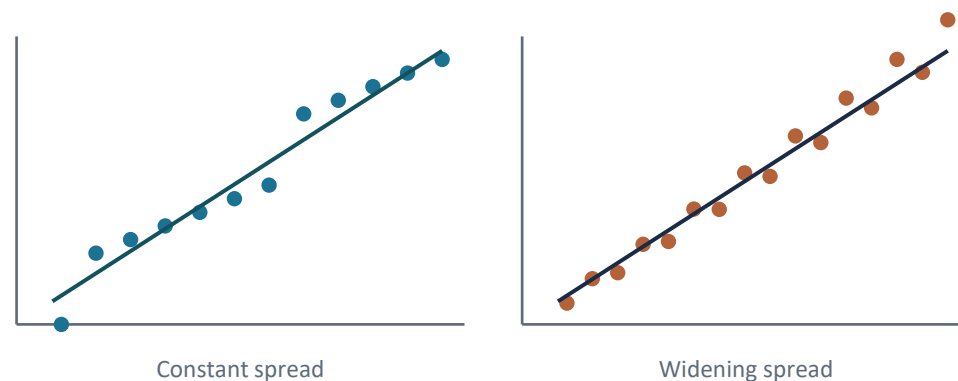
Error terms should not depend on one another — as they often do in time-series data, where each value depends on the previous one.



A wave-like or trending pattern in the residuals is a red flag that observations are not independent.

HOMOSCEDASTICITY

Error terms should have constant variance — the spread should not grow or shrink as the predicted value changes.



A cone-shaped spread (heteroscedasticity) means predictions are less reliable at some ranges than others.

Assumptions III: Multicollinearity & Autocorrelation

NO OR LITTLE MULTICOLLINEARITY

Multicollinearity occurs when the independent variables are too highly correlated with each other.

Example: including both “height in cm” and “height in inches” as separate features — they carry the same information, which confuses the model about which one truly matters.

NO AUTOCORRELATION

Autocorrelation occurs when residuals are not independent from each other across observations.

Example: stock prices, where today's price is not independent of yesterday's price — a common issue in time-series data.

Multicollinearity: VIF & Consequences

VARIANCE INFLATION FACTOR (VIF)

VIF = 1

No correlation with other predictors

$1 < \text{VIF} < 5$

Moderate correlation — generally acceptable

$5 \leq \text{VIF} < 10$

High correlation — worth investigating

$\text{VIF} \geq 10$

Severe multicollinearity — action needed

Tolerance = $1/\text{VIF}$ — a tolerance below 0.1 corresponds to VIF above 10.

IF MULTICOLLINEARITY IS PRESENT

- Coefficient estimates become unstable and can flip sign with small data changes
- Standard errors inflate, widening confidence intervals and shrinking t-statistics
- Individual coefficients lose their “holding all else constant” interpretation
- Overall model fit and prediction accuracy are usually unaffected
- Fixes: drop a redundant variable, combine features, use PCA, or switch to Ridge regression

How to Check Each Assumption

Each assumption has a standard diagnostic and an accepted threshold used to judge whether it holds.

Assumption	How to Check It	Acceptable Value
Linearity	Scatter plot of X vs Y, or Residuals vs Fitted plot	No visible curve — points scattered evenly around a flat, straight band
Normality	Q-Q plot of residuals; Shapiro-Wilk or Jarque-Bera test	Q-Q points fall near the diagonal; test p-value > 0.05 (fail to reject normality)
Independence	Durbin-Watson test on residuals	Statistic close to 2 (commonly 1.5 to 2.5 is considered acceptable)
Homoscedasticity	Residuals vs Fitted plot; Breusch-Pagan test	No funnel/cone shape; test p-value > 0.05 (fail to reject constant variance)
No Multicollinearity	Variance Inflation Factor (VIF); correlation matrix	VIF < 5 (some practitioners allow up to 10); pairwise correlation below ~0.8
No Autocorrelation	Durbin-Watson test (same test as Independence)	Statistic close to 2; well below 2 signals positive autocorrelation, well above signals negative

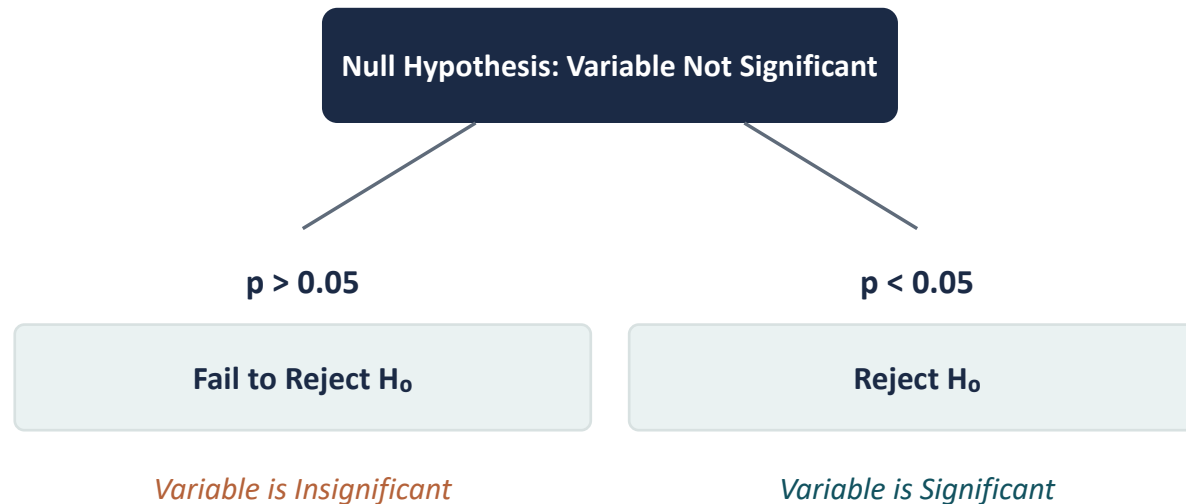
These tests are commonly run using statsmodels in Python (e.g. statsmodels.stats.diagnostic, variance_inflation_factor, durbin_watson) — the same library used for the p-value table in the notebook.

Multiple Linear Regression

Multiple Linear Regression extends the same idea to several input features at once, finding the best-fit equation across all of them together.

$$\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \dots + \beta_nx_n$$

Each coefficient is tested individually with a p-value, to check whether that feature genuinely helps the model or could be dropped.



Hypothesis Testing: Is the Line Significant?

To know whether the fitted line is meaningful, we test whether its slope (β_1) is actually different from zero.

Null Hypothesis (H_0): $\beta_1 = 0$

Alternate Hypothesis (H_a): $\beta_1 \neq 0$

FAIL TO REJECT H_0

β_1 is effectively zero — the feature is insignificant and of no real use in the model.

REJECT H_0

β_1 is not zero — the fitted line represents a statistically significant relationship.

P-VALUE IN HYPOTHESIS TESTING

The p-value helps us decide whether we have enough evidence in the data to reject the Null Hypothesis (H_0).

1. HYPOTHESIS SETUP



- **Null Hypothesis (H_0):** Assumes no effect or no difference.
- **Alternative Hypothesis (H_1):** Assumes an effect or difference exists.
- We collect sample data and calculate a test statistic.

2. WHAT IS P-VALUE?



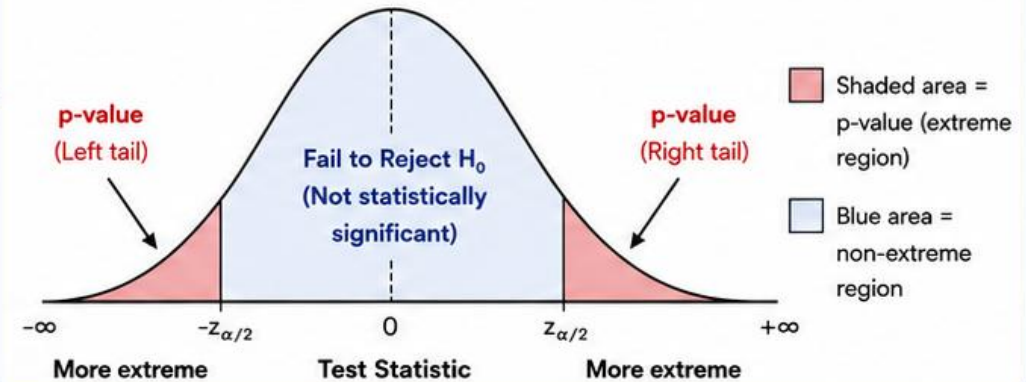
The p-value is the **probability** of observing data as extreme as (or more extreme than) what we observed, **assuming the Null Hypothesis (H_0) is true**.



Smaller p-value → Stronger evidence against H_0

3. VISUAL UNDERSTANDING

Sampling distribution of the test statistic under H_0



4. DECISION RULE

Compare p-value with Significance Level (α)

- α (alpha) is the threshold for deciding statistical significance.
- Common α values: 0.10, 0.05, 0.01

If p-value $\leq \alpha \rightarrow$ **Reject H_0**

If p-value $> \alpha \rightarrow$ **Fail to Reject H_0**

5. INTERPRETATION GUIDE

P-VALUE	INTERPRETATION	CONCLUSION (AT $\alpha = 0.05$)
p-value ≤ 0.01	 Very strong evidence against H_0	Reject H_0 (Highly significant)
$0.01 < \text{p-value} \leq 0.05$	 Moderate evidence against H_0	Reject H_0 (Statistically significant)
$0.05 < \text{p-value} \leq 0.10$	 Weak evidence against H_0	Fail to Reject H_0 (Not significant)
p-value > 0.10	 Little or no evidence against H_0	Fail to Reject H_0 (Not significant)



KEY TAKEAWAY

The p-value tells us how surprising our data is under the assumption that H_0 is true.

Smaller p-value = More surprising data = Stronger evidence to reject H_0 .

EXAMPLE (TWO-TAILED TEST)

- Test statistic (z) = 2.35
- p-value = 0.018
- At $\alpha = 0.05$

Since p-value (0.018) ≤ 0.05
 $\rightarrow$ **Reject H_0**

Worked Example: Predicting House Price

Larger area tends to mean a higher price — a roughly linear relationship. We fit a line that best represents this trend.

Area (sq.ft)	500	750	1000	1250	1500
Price (Lakhs)	1.8	2.6	3.4	4.2	5.1

FITTED MODEL FROM THIS DATA

Price ≈ 0.0033 × Area + 0.14

For a 2000 sq.ft house, the model predicts a price of roughly 6.7 lakhs — the same formula generalizes beyond the training rows above.



FITTED MODEL

$\hat{y} = 0.0033x + 0.14$

where,
x = Area (sq.ft)
 $\hat{y}$ = Predicted Price (Lakhs)

INTERPRETATION

- ✓ Larger house area generally results in a higher price.
- ✓ The regression line captures the overall linear trend.
- ✓ New house prices are predicted using the fitted equation.

EXAMPLE PREDICTION

For a 2000 sq.ft house:
 $\hat{y} = 0.0033(2000) + 0.14$
 $= 6.60 + 0.14$
 $\approx 6.7 \text{ Lakhs}$

Where Linear Regression Is Used

1

House Price Prediction

Estimate price from area, location, and number of rooms.

2

Sales Forecasting

Predict next month's sales from historical trends.

3

Salary Estimation

Predict salary from years of experience and role.

4

Temperature Prediction

Estimate temperature from time of year and location.

HANDS-ON · PRACTICAL 2

Linear Regression Demo

Open the notebook: train a Linear Regression model and evaluate it using RMSE and R^2 .

HANDS-ON PRACTICAL



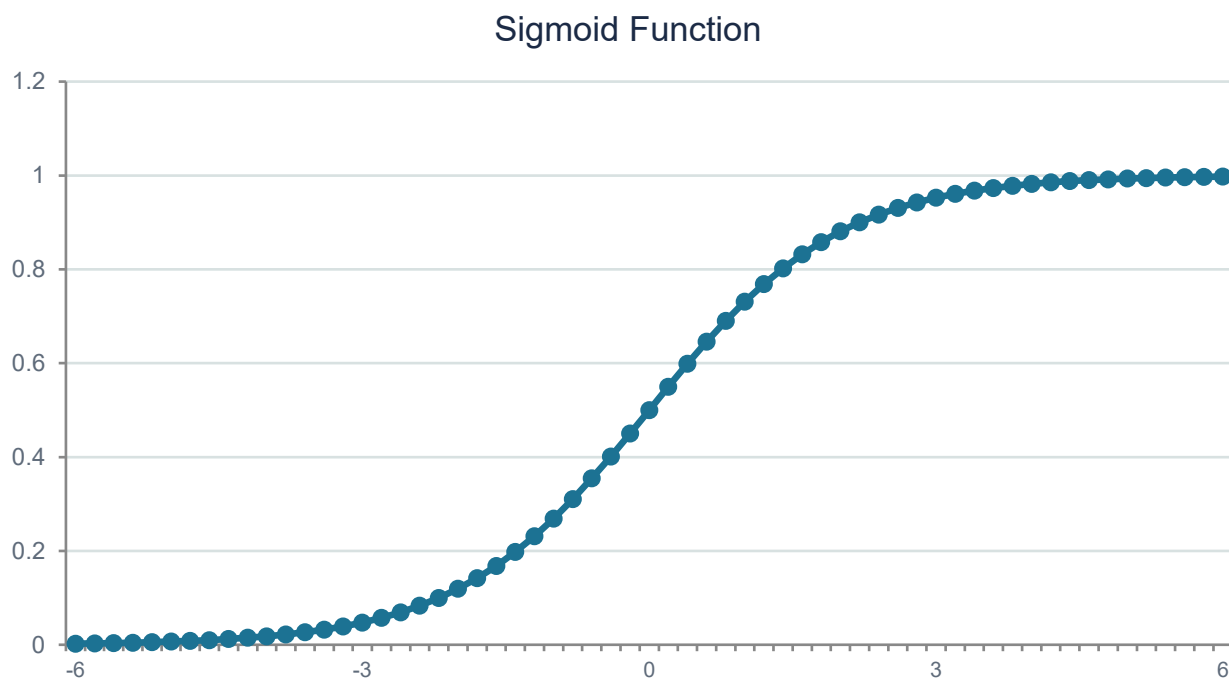
PART 2 · CLASSIFICATION

Logistic Regression



From a Line to a Probability

Logistic Regression is used for classification, not for predicting numbers. It squeezes any input into a probability between 0 and 1 using the sigmoid function.



- Output near 0: the model predicts “No.”
- Output near 1: the model predicts “Yes.”
- A threshold (usually 0.5) decides the final class.

LOGISTIC REGRESSION

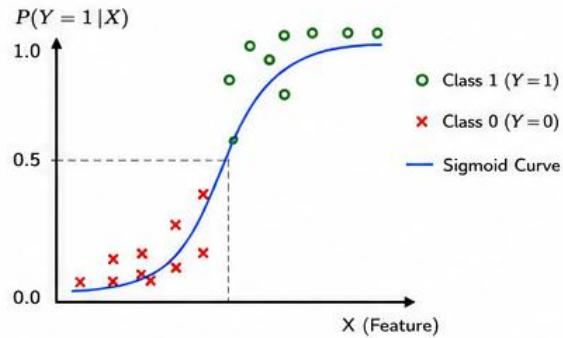
A **supervised** machine learning algorithm used for classification problems.
It predicts the probability of a binary (categorical) dependent variable ($Y \in \{0, 1\}$).

1 What is Logistic Regression?

It models the relationship between independent features (X) and a binary dependent variable (Y) by estimating the **probability** that $Y = 1$.

Instead of fitting a straight line, it fits an S-shaped curve (Sigmoid) that outputs probabilities between 0 and 1.

2 Binary Classification Visualization

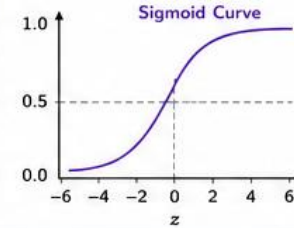


3 Sigmoid Function

Logistic Regression uses the **Sigmoid** (Logistic) function to map any real-valued number into a probability between 0 and 1.

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \text{ where } z = \beta^T x + \beta_0$$

z	$\sigma(z)$
$-\infty$	0
-4	0.018
-2	0.119
0	0.500
2	0.881
4	0.982
$+\infty$	1



4 Model / Equation

The probability that $Y = 1$ given X is modeled as,

$$P(Y = 1 | X) = \sigma(z) = \frac{1}{1 + e^{-(\beta^T x + \beta_0)}}$$

where,

$X = [x_1, x_2, \dots, x_n]$: Feature vector

$\beta = [\beta_1, \beta_2, \dots, \beta_n]$: Coefficient vector

β_0 : Intercept (bias)

$z = \beta^T x + \beta_0$: Linear combination

Prediction:

If $P(Y = 1 | X) \geq 0.5 \rightarrow$ Predict 1 (Class 1)
Else $\rightarrow$ Predict 0 (Class 0)

5 Cost Function (Log Loss)

Logistic Regression uses **Log Loss** to measure the error.
For m training examples:

$$J(\beta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(p^{(i)}) + (1 - y^{(i)}) \log(1 - p^{(i)})]$$

where,

$p^{(i)} = P(Y = 1 | X^{(i)})$ (Predicted probability)

$y^{(i)} \in \{0, 1\}$ (Actual label)

Goal: Find β that minimize $J(\beta)$.

6 Assumptions

- Observations are independent.
- No perfect multicollinearity among features.
- Linearity of independent variables and log-odds.
- Large sample size gives better results.
- The dependent variable is binary.



Logistic Regression models the log-odds as a linear combination of features.

7 Coefficients Interpretation

The coefficients (β) represent the change in log-odds of the outcome for a one-unit increase in the feature.

$$\text{log-odds} = \log\left(\frac{P(Y = 1 | X)}{1 - P(Y = 1 | X)}\right) = \beta^T x + \beta_0$$

If $\beta_j > 0 \rightarrow$ Increasing x_j increases the probability of class 1.

If $\beta_j < 0 \rightarrow$ Increasing x_j decreases the probability of class 1.

8 Pros and Cons



Pros

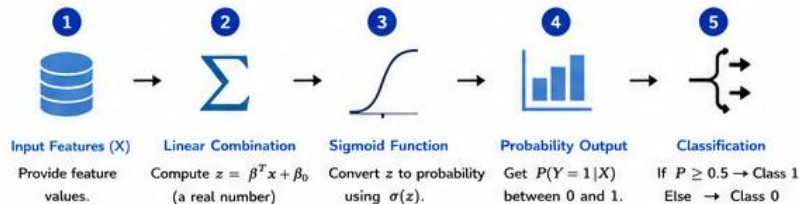
- ✓ Simple and easy to implement
- ✓ Interpretable (coefficients show impact on log-odds)
- ✓ Works well for linearly separable data
- ✓ Large probability as output (0 to 1)



Cons

- ✗ Assumes linear relationship between features and log-odds
- ✗ Sensitive to outliers
- ✗ Not suitable for complex non-linear patterns
- ✗ Performance depends on feature engineering

9 How it works (Step by Step)

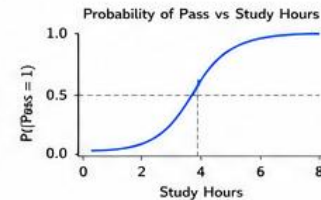


10 Example (Small Dataset)

Study Hours (X)	Pass (1) / Fail (0) (Y)
1	0
2	0
3	0
4	1
5	1
6	1
7	1

Interpretation:

- When study hours $< 4 \rightarrow$ Probability of pass is low.
- When study hours $> 4 \rightarrow$ Probability of pass is high.
- At 4 hours $\rightarrow$ Probability is around 0.5 (50%).



11 Evaluation Metrics

Common metrics for binary classification:

Accuracy
Overall correctness

Precision
Correct positive predictions

Recall
Capture all positives

F1 Score
Balance between precision & recall

Confusion Matrix

Actual \ Predicted	1 (Positive)	0 (Negative)
1 (Positive)	TP	FN
0 (Negative)	FP	TN

TP: True Positive
FP: False Positive
FN: False Negative
TN: True Negative



Note: Logistic Regression is used for binary classification problems.

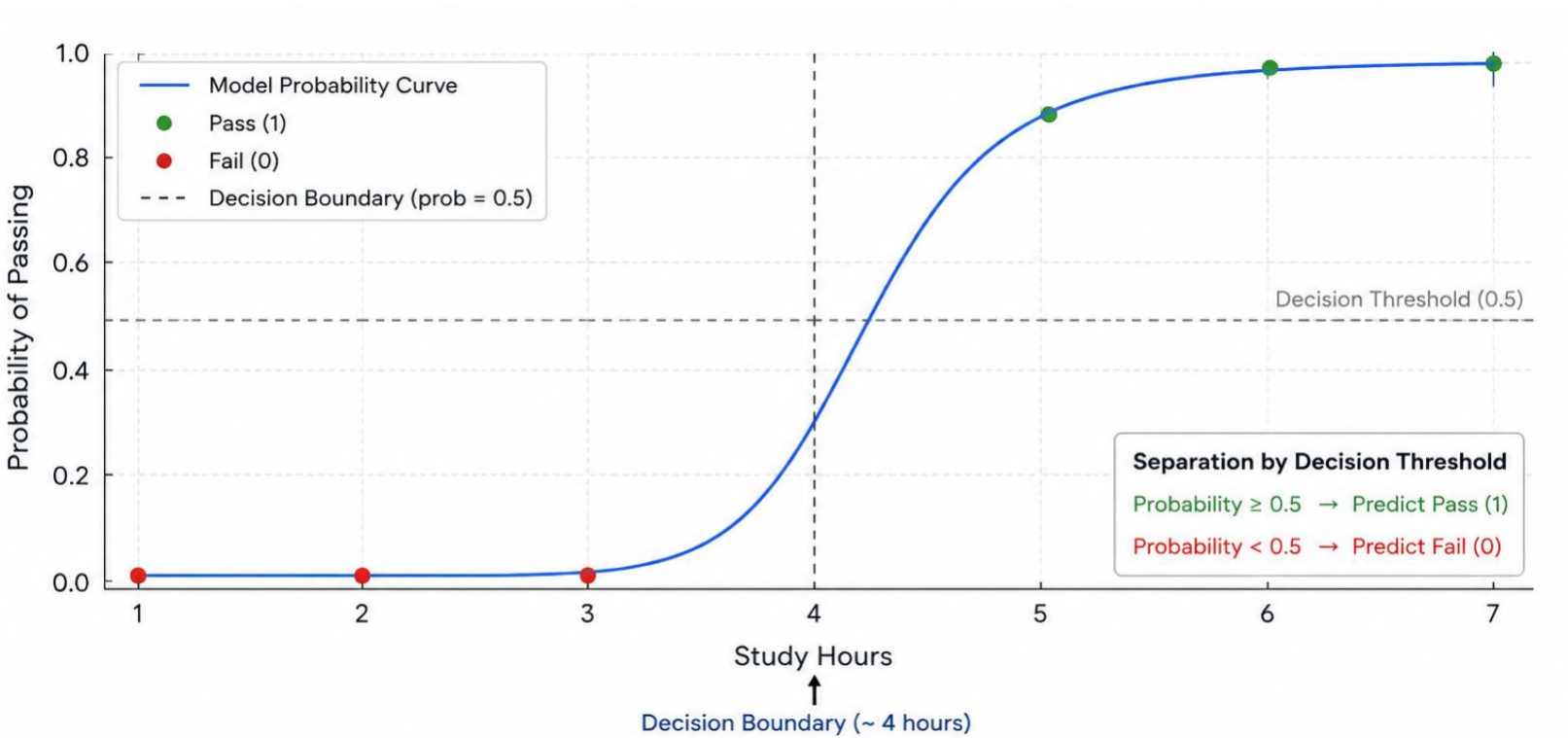
For multi-class classification, use **Softmax Regression** (Multinomial Logistic Regression).



Worked Example: Predicting Exam Pass/Fail

A small dataset of study hours and pass/fail outcomes, and the probability curve the model fits to it.

Study Hours	1	2	3	4	5	6	7
Pass (1) / Fail (0)	0	0	0	0	1	1	1



INTERPRETATION

- When study hours are below 4, the predicted probability of passing is low.
- When study hours are above 4, the predicted probability of passing is high.
- At around 4 hours, the predicted probability crosses 0.5 — the model's decision threshold.

The Decision Boundary

Logistic Regression separates classes with a straight boundary line. Points on either side are classified differently.



HANDS-ON · PRACTICAL 3

Logistic Regression Demo

Open the notebook: train a classifier and evaluate with accuracy, precision, recall, F1, and the confusion matrix.

HANDS-ON PRACTICAL



Key Takeaways From Today

- Machine Learning learns rules from data instead of being told every rule, and splits into supervised, unsupervised, and reinforcement learning.
- Always split data into training and testing parts before evaluating a model.
- Evaluation metrics (accuracy, precision, recall, F1, RMSE, R^2) tell us how good a model really is.
- Linear Regression predicts continuous numbers using a best-fit line.
- Logistic Regression predicts classes using a probability from the sigmoid function.

Thank You

Questions & Discussion

Next up: Basic Machine Learning — Session II (Decision Trees, Random Forest & more)

